

Dear Vietnam Summer School of Science (VSSS) 2026,

I did not come to science through a straight path. My academic journey has been more like an adaptive control process: learning from errors, adjusting by feedbacks, and gradually finding stability. For this reason, I see that the theme of VSSS 2026 “**Rethinking Science**” reflects the period I am currently going through, looking back at how I understand science, growing through each academic experience, and preparing more seriously for a long-term academic path.

My interest in science and technology started early, through some scientific competitions during my school years. At that time, what attracted me most was the feeling of turning an idea in my mind into a working model that could be presented in a competition. I used to think that doing science mainly meant developing or improving existing research topics. But the further I went, the more I realized that research requires a deeper question: How can I develop a research topic in an academic way and make a real contribution to the community?

When I entered university and joined a research lab early, I had the chance to join in different projects, ranging from electronics, artificial intelligence to robotics, and control. These experiences helped me build a solid technical foundation and enhance my research skills. At the same time, they made me ask whether I was truly developing a strong research direction or simply moving through many separate experiences. In my final year, this question actually became clearer. I realized that a person who wants to follow an academic path cannot only ask, “What can I build?” I also need to ask what the problem means, where its current limits are, and whether I have enough patience and depth to pursue it seriously.

Initially, this doubt made me feel really uncertain but later, I came to see it as an useful feedback signal. In a control system, an error does not only show that the system has not reached stability. If observed properly, it helps the system adjust itself. For me, the process of rethinking science also began from recognizing the gap between my desire to do research and my actual ability, the difference between completing a project and asking a meaningful research question, and the tension between being interested in many fields and needing to build a clearer academic direction.

To test this direction more seriously, I decided to take a gap year and step outside my comfort zone. I wanted to place myself in different academic and research environments, gain new experiences, rethink the way I understood research, and also prepare for my graduate studies.

During my research experience in Taiwan on tool wear prediction in CNC machining, I saw the large gap between a model and a real working system. Machining data is not always clean, measurement signals are also not always ideal, and a model is only meaningful when it can help solve problems in the system efficiency and reliability. From that experience, research was no longer just a modeling problem to me. It became an effort to answer clearer questions, “What problems does this research solve? Who does it serve? And in what context does it matter?”

Later, in Singapore, while working on underwater wireless power transfer (UWPT) systems for underwater robot, I received another important feedback signal. I realized that modern robotic systems cannot be understood through a single field alone. In fact, a complete AUV model, mechanical design, control, and energy are connected together. The problem of UWPT also showed me a concrete technical challenge for maintaining the efficiency and stability under position errors, weak coupling, and environmental uncertainties. This experience helped me define my current research interests more clearly on underwater robotics, adaptive control, and wireless power transfer for autonomous systems.

Reflecting back, I am grateful for those unstable periods. They did not make me lose my interest in science. Instead, they forced me to think more maturely about how I do science. I gradually understood that research does not begin with trying to create something new at any cost. It begins with having a research foundation, understanding a meaningful problem, seeing the limits of current

approaches, asking a clear question, and building a clear research path. For me, this is an important change from the mindset of doing projects to the mindset of building a clear research direction.

Along this journey, I had opportunity to work closely with VSSS alumni, sharing with me their VSSS experiences helped me understand the spirit and value of the program more clearly. Through their sharing, I saw VSSS as a strong academic network where young researchers can connect and support each other to enhance the value of Vietnamese researchers in the world. Their stories became an important motivation for me to apply to VSSS again this year.

I hope to join VSSS 2026 to listen to the scientific stories of leading professors and young researchers from different backgrounds. With my own experiences in different research environments, I hope to contribute a sincere and practical perspective to the discussions at VSSS. Like one pendulum in a larger system, I wish to become part of this academic network, learning from others while also contributing my own small movement to its shared values.

Sincerely,

Long Nguyen Khac